Project Design Phase
Solution Architecture

Date	8 March 2026
Team ID	69aa7af9d483472145093618
Project Name	Libspace
Maximum Marks	4 Marks

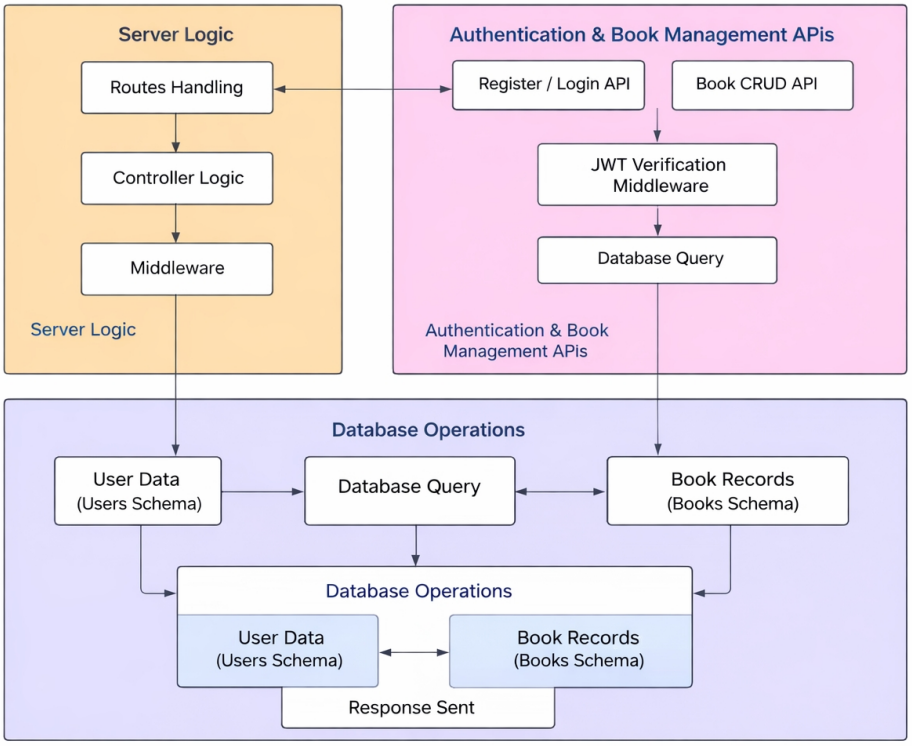
Solution Architecture:

The solution architecture for Libspace focuses on building a secure, scalable, and maintainable backend system using Node.js, Express.js, and MongoDB. The backend exposes RESTful APIs for user authentication and book management while ensuring strict data privacy through JWT-based authorization.

The architecture is designed to maintain clear separation of concerns between authentication, business logic, and database layers, ensuring a clean and modular backend structure.

Goals of the Solution Architecture:

- **Identify the Best Tech Solution:**
  Utilize Node.js, Express.js, MongoDB, JWT, and bcrypt to build a secure and efficient backend API for book management.
- **Define Structure & Characteristics:**
  Follow a modular MVC architecture to ensure scalability, maintainability, and clear separation between authentication and book management logic.
- **Outline Features & Development Phases:**
  Define development milestones including backend setup, authentication APIs, book CRUD operations, middleware protection, and API testing.
- **Establish Specifications for Development & Delivery:**
  Implement RESTful API standards, JWT-based authorization, structured middleware, and proper HTTP status handling to ensure secure and consistent data flow.
- **Ensure Performance & Scalability:**
  Optimize database queries and asynchronous API handling to provide fast response times and support future feature expansion.


Server Logic
Routes Handling
Controller Logic
Middleware
Server Logic
Authentication & Book Management APIs
Register / Login API
Book CRUD API
JWT Verification Middleware
Database Query
Authentication & Book Management APIs
Database Operations
User Data (Users Schema)
Database Query
Book Records (Books Schema)
Database Operations
User Data (Users Schema)
Book Records (Books Schema)
Response Sent